

Muscular Dystrophy, Congenital Hearing Loss, and Ovarian Insufficiency Syndrome (MDHLOV): A Comprehensive Disease Report

Disease: Muscular Dystrophy, Congenital Hearing Loss, and Ovarian Insufficiency Syndrome
MONDO ID: MONDO:0859189 · **OMIM:** #619518 (MDHLOV) · **Gene:** *GGPS1* (1q42.3) **Category:** Mendelian (autosomal recessive)

Summary

Muscular Dystrophy, Congenital Hearing Loss, and Ovarian Insufficiency Syndrome (MDHLOV; also abbreviated MDHLO/MUDHLOV) is an **ultra-rare autosomal-recessive multisystem disorder** caused by biallelic hypomorphic missense variants in *GGPS1*, the gene on chromosome 1q42.3 encoding **geranylgeranyl diphosphate synthase (GGPPS/GGDPS)**. This enzyme sits in the mevalonate/isoprenoid pathway and produces geranylgeranyl pyrophosphate (GGPP, C20), the lipid donor used to geranylgeranilate small GTPases (the Rab and Rho families). The syndrome was first defined by Foley et al. in 2020, who identified 5 different biallelic pathogenic *GGPS1* variants in 11 patients from 6 families, and has since been expanded by additional cohorts.

The core clinical picture is a **fully penetrant, congenital-onset, progressive proximal muscular dystrophy** (elevated creatine kinase; dystrophic histology with centrally nucleated fibers and rimmed vacuoles) accompanied by two variable extra-muscular features: **congenital sensorineural hearing loss** and, in post-pubertal females, **primary/premature ovarian insufficiency**. The combination of sensorineural hearing loss plus ovarian insufficiency overlaps clinically with **Perrault syndrome**, and *GGPS1* is now counted among the Perrault-syndrome gene set. Later reports have shown the two non-muscle features are

variable — hearing loss was present in only 46% of one cohort, and at least one patient had isolated proximal weakness with hepatic transaminase elevation and spared hearing — expanding the phenotypic spectrum.

Mechanistically, reduced GGPP supply is inferred to impair prenylation-dependent membrane anchoring of small GTPases (particularly Rab proteins), degrading vesicle trafficking and the Ca^{2+} -dependent sarcolemmal membrane-repair machinery. Direct functional support comes from **delayed membrane healing after laser injury in patient-derived myogenic cells** and a disease-specific **Y259C knock-in mouse**. There is **no disease-specific therapy**; management is supportive (respiratory care, hearing rehabilitation, hormone replacement for ovarian insufficiency, physical therapy, orthopedic management of scoliosis). This report consolidates 10 confirmed findings and 25 reviewed papers across all 15 requested sections.

Key Findings

Finding 1 — Biallelic *GGPS1* variants cause the muscle-ear-ovary triad

The foundational discovery (Foley et al., 2020) used **whole exome sequencing superimposed on shared-haplotype mapping** to identify biallelic *GGPS1* variants as the cause of this syndrome. Eleven patients across six families carried five distinct biallelic pathogenic variants. The gene product, **geranylgeranyl diphosphate synthase**, catalyzes the synthesis of geranylgeranyl pyrophosphate, the lipid precursor of geranylgeranylated proteins including small GTPases. The clinical triad was: proximal muscular dystrophy in all 11 patients (100%), congenital sensorineural hearing loss in 10 of 11, and primary ovarian insufficiency in all post-pubertal females. Inheritance is autosomal recessive.

"We applied whole exome sequencing (WES) superimposed on shared haplotype regions to identify the initial biallelic variants in *GGPS1*." — [PMID: 32403198](#) "In addition to proximal weakness, all but one patient presented with congenital sensorineural hearing loss, and all postpubertal females had primary ovarian insufficiency." — [PMID: 32403198](#)

Finding 2 — Hearing loss and ovarian insufficiency are variable; the spectrum includes hepatic involvement

Two subsequent reports broadened the phenotype. Kaiyrzhanov et al. (2022) described 11 additional individuals from 4 families with missense *GGPS1* variants in whom **hearing loss was present in only 46%**, and concluded the data "demonstrates that hearing loss and ovarian insufficiency might be a variable feature of the GGPS1-associated muscular dystrophy." Altassan et al. (2024) reported a patient who "presented with only proximal muscle weakness, and elevated liver transaminases with spared hearing function," adding **hepatic involvement** to the recognized spectrum and confirming that hearing can be spared.

"hearing loss was present in only 46% of the individuals ... demonstrates that hearing loss and ovarian insufficiency might be a variable feature of the GGPS1-associated muscular dystrophy." — [PMID: 35869884](#) "The patient presented with only proximal muscle weakness, and elevated liver transaminases with spared hearing function." — [PMID: 38129970](#)

Interpretation: Myopathy is the obligate, fully penetrant feature; sensorineural hearing loss and ovarian insufficiency show variable expressivity. This variable expressivity is why the disorder is sometimes ascertained as "Perrault-syndrome-like" and sometimes as an isolated congenital muscular dystrophy.

Finding 3 — Mechanism: reduced GGPP impairs Rab/small-GTPase geranylgeranylation and sarcolemmal membrane repair

Foley et al. demonstrated **delayed membrane healing after laser injury in patient-derived myogenic cells** and generated a **Y259C knock-in mouse**, whose muscle histology was dystrophic with ultrastructural autophagic material and enlarged mitochondria. GGPP is the obligate lipid precursor for geranylgeranylation of small GTPases including the Rab family. Independent literature establishes the mechanistic links: Ca^{2+} -dependent vesicle-fusion-based sarcolemmal resealing is an active, essential process in skeletal muscle and is defective in membrane-repair myopathies (Bansal et al., 2003); and Rab GTPases require C-terminal prenylation for membrane tethering. Removing the geranylgeranyl lipid group therefore strips Rabs of their ability to anchor to membranes and drive vesicle trafficking.

"There was delayed membrane healing after laser injury in patient-derived myogenic cells." — PMID: 32403198 "Membrane repair is therefore an active process in skeletal muscle fibres." — PMID: 12736685 "Rabs oscillate between an inactive GDP-bound conformation and an active GTP-bound state that is tethered to lipid membranes via a C-terminal prenylation site on conserved cysteine residues." — PMID: 23663983

Finding 4 — Identifiers and constraint: *GGPS1* is not LoF-constrained, consistent with recessive hypomorphic disease

Attribute	Value
Disease MONDO	MONDO:0859189
Disease OMIM	#619518 (MDHLOV)
MedGen / UMLS	C5561980 / 1794190
Gene HGNC	HGNC:4249
NCBI Gene	9453
Ensembl	ENSG00000152904
Cytoband	1q42.3 (GRCh38 chr1:235,327,350–235,344,544)
Gene MIM	606982
RefSeq transcript	NM_004837.4
Aliases	MDHLO, MUDHLOV

gnomAD constraint metrics show **pLI = 0.007**, observed/expected LoF = 0.53 (90% CI 0.36–0.82), and missense Z = 1.47 — i.e., *GGPS1* is **not** strongly haploinsufficient or LoF-intolerant. This is fully consistent with an autosomal-recessive mechanism in which biallelic **hypomorphic missense** alleles (partial loss of function), rather than complete nulls, cause disease. Complete loss of GGPP synthesis is presumed incompatible with life (see Finding 9).

Finding 5 — Pathogenic missense variants cluster in the C-terminal catalytic/substrate-binding region

ClinVar (NM_004837.4) pathogenic/likely-pathogenic *GGPS1* SNVs cluster tightly:

cDNA	Protein	ClinVar significance	Notes
c.776A>G	p.Tyr259Cys	Pathogenic	Used for the knock-in mouse
c.781C>G	p.Arg261Gly	Pathogenic	Most recurrent allele (mutational hotspot)
c.782G>A	p.Arg261His	Conflicting	Same residue hotspot
c.770T>G	p.Phe257Cys	Likely pathogenic	
c.854T>G	p.Val285Gly	Pathogenic	
c.764G>A	p.Gly255Asp	VUS	
c.790C>G	p.Leu264Val	VUS	
c.545T>C	p.Leu182Pro	VUS	

All disease-causing variants are **ultra-rare missense substitutions** falling in specific catalytic domains of the enzyme, with **Arg261 an apparent mutational hotspot**. Large 1q42 copy-number variants in ClinVar are non-specific and not associated with this Mendelian disorder.

"Ultra-rare biallelic pathogenic variants in geranylgeranyl diphosphate synthase 1 (GGPS1) have recently been associated with muscular dystrophy/hearing loss/ovarian insufficiency syndrome." — [PMID: 35869884](#) "A total of 11 patients in 6 families carrying 5 different biallelic pathogenic variants in specific domains of GGPS1 were identified." — [PMID: 32403198](#)

Finding 6 — Pathogenic alleles are absent-to-ultra-rare in gnomAD v4

Variant	Protein	gnomAD v4 alleles	Approx. AF	Homozygotes
c.776A>G	p.Tyr259Cys	0	absent	0
c.854T>G	p.Val285Gly	0	absent	0
c.781C>G	p.Arg261Gly	34	$\sim 2.2 \times 10^{-5}$	0
c.782G>A	p.Arg261His	~	$\sim 1.0 \times 10^{-5}$	0
c.545T>C	p.Leu182Pro (VUS)	8	$\sim 4.7 \times 10^{-6}$	0
c.764G>A	p.Gly255Asp (VUS)	1	—	0

No **homozygotes** are reported for any pathogenic allele, consistent with recessive selection against homozygous carriers. p.Arg261Gly is the most recurrent pathogenic allele.

Finding 7 — HPO phenotype spectrum with frequencies

Curated HPO annotations (OMIM:619518; n $\approx$ 11 patients):

Phenotype	HPO term	Frequency
Progressive muscle weakness	HP:0003323	11/11 (100%)
Congenital onset	HP:0003577	11/11 (100%)
Elevated serum creatine kinase	HP:0003236	9/9 (100%)
Centrally nucleated skeletal muscle fibers	HP:0003687	9/9
Rimmed vacuoles	HP:0003805	9/9
Skeletal muscle autophagosome accumulation	HP:0025717	2/9
Mitochondrial hypertrophy	HP:0033686	1/9
Sensorineural hearing impairment	HP:0000407	10/11 (91%)*
Premature ovarian insufficiency	HP:0008209	3/3
Female infertility	HP:0008222	3/3
Short stature	HP:0004322	8/11
Failure to thrive	HP:0001508	7/10
Respiratory insufficiency	HP:0002093	8/10
Reduced forced vital capacity	HP:0032341	2/4
Loss of ambulation	HP:0002505	5/11
Scoliosis	HP:0002650	4/10
Motor delay	HP:0001270	—
Decreased fetal movement	HP:0001558	—
Poor suck	HP:0002033	—
Weak cry	HP:0001612	—
Autosomal recessive inheritance	HP:0000007	—

*The 91% figure derives from the original Foley cohort; the pooled frequency across later cohorts is lower (~46% in Kaiyrzhanov et al.), reflecting variable expressivity.

Finding 8 — GGPS1 protein: cytosolic prenyltransferase homoheamer localized to the sarcomere Z-line

UniProt **O95749** (human GGPS1, 300 aa) catalyzes the sequential trans-addition of isopentenyl diphosphate (IPP) onto dimethylallyl/geranyl/farnesyl diphosphate to form GGPP (C20). Its quaternary structure is a **homoheamer** (a trimer of homodimers), consistent with crystal structures of bacterial GGPP synthases (e.g., *Nonlabens dokdonensis*, which "forms a hexamer composed of homodimeric trimer"). Subcellular localization spans the cytoplasm, perinuclear region, and — notably for muscle disease — the **myofibril/sarcomere Z-line**. The UniProt disease annotation (MDHLO) reads: "An autosomal recessive disorder characterized by early-onset progressive muscle weakness, sensorineural hearing loss, and primary amenorrhea due to ovarian insufficiency. Some patients become wheelchair-bound by the second decade, whereas others have a milder phenotype and maintain independent ambulation."

"GGPS1 encodes geranylgeranyl diphosphate synthase in the mevalonate/isoprenoid pathway, which catalyzes the synthesis of geranylgeranyl pyrophosphate, the lipid precursor of geranylgeranylated proteins including small guanosine triphosphatases." — [PMID: 32403198](#)

Finding 9 — Model organisms and evolutionary conservation

GGPS1 is highly conserved with clear orthologs across the tree of life: mouse *Ggps1* (NCBI Gene 14593; MGI), rat *Ggps1* (291211), zebrafish *ggps1* (336798), *Xenopus ggps1* (549876), *Drosophila* (NCBI 38816), and *S. cerevisiae* **BTS1** (856036). Foley et al. generated a disease-specific **p.Tyr259Cys knock-in mouse**. Independent mouse work shows *Ggpps* has essential roles: skeletal-muscle-specific deletion causes insulin resistance via RhoA geranylgeranylation ([PMID: 26112408](#)), and liver-specific deletion alters adipose remodeling via Rab27A-dependent extracellular-vesicle secretion ([PMID: 32024826](#)). Global loss of *Ggpps* is not viable — consistent with GGPP being an essential isoprenoid and with the human disease requiring hypomorphic (not null) alleles.

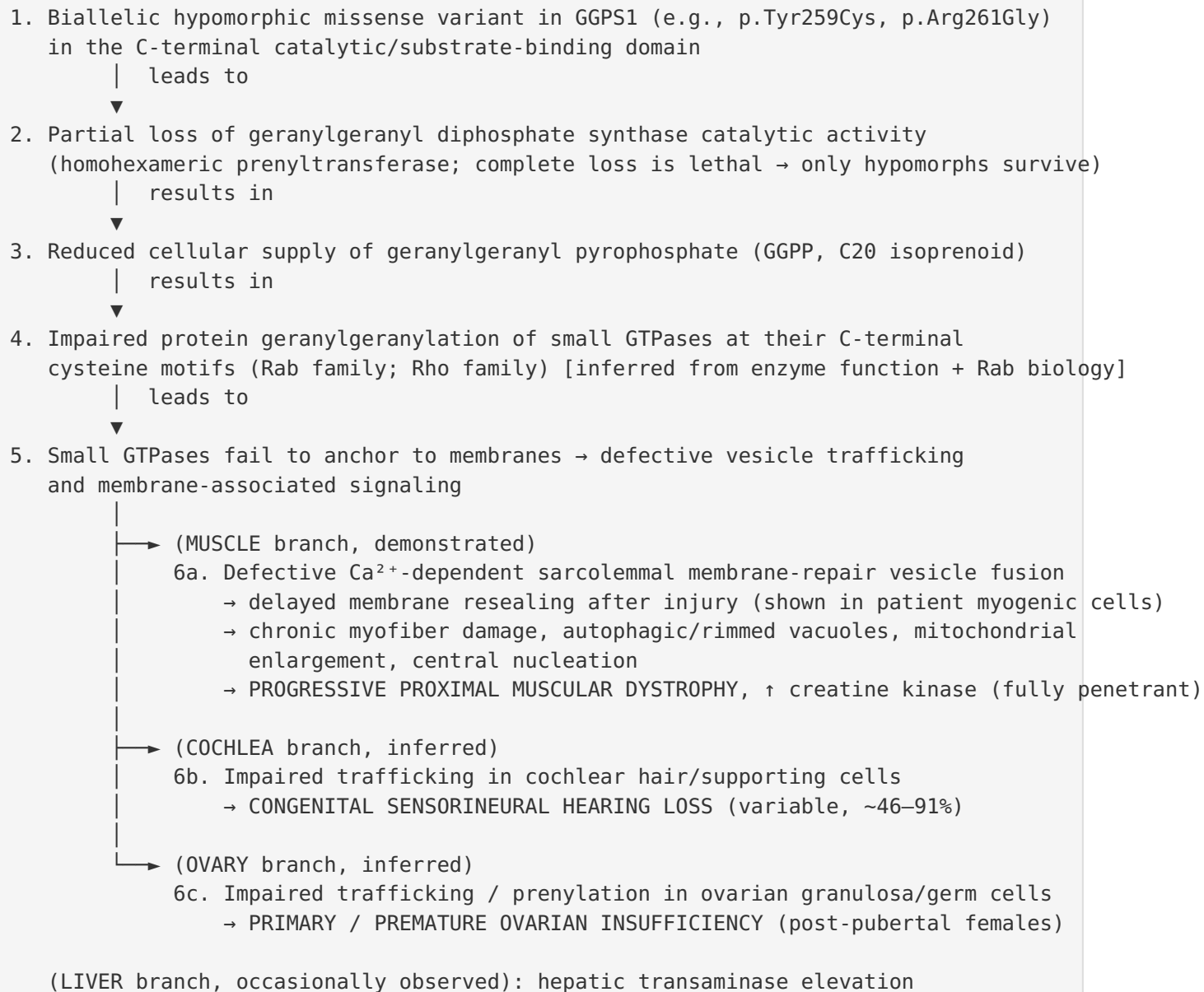
"the generation of a Y259C knock-in mouse were done." — [PMID: 32403198](#) "we generated mice with specific GGPPS deletions in their skeletal muscle tissue." — [PMID: 26112408](#)

Finding 10 — GGPS1 belongs to the polyprenyl-synthetase fold; disease residues line the catalytic domain

InterPro annotation of O95749: Pfam **PF00348** (polyprenyl synthetase domain); InterPro IPR000092 (polyprenyl synthetase-like family), IPR008949 (isoprenoid synthase domain superfamily), IPR033749 (conserved site); CDD cd00685 (trans-isoprenyl diphosphate synthases, head-to-tail); CATH G3DSA:1.10.600.10 (farnesyl-diphosphate-synthase homologous superfamily); PROSITE PS00444/PS00723; PANTHER PTHR12001 (GGPP synthase family). The pathogenic residues **Phe257**, **Tyr259**, **Arg261**, and **Val285** all fall within this C-terminal catalytic/substrate-binding domain, providing a structural rationale for their hypomorphic effect on enzyme activity.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)



Upstream vs downstream. The upstream lesion is the enzymatic deficit (steps 1–3); the downstream effectors are the under-prenylated small GTPases and the trafficking/repair failures they cause (steps 4–6). The muscle branch is the best-supported (direct patient-cell and mouse evidence); the cochlear and ovarian branches are mechanistically inferred by analogy to the same prenylation defect acting in tissues with high secretory/trafficking demand.

Pathways, cell types, compartments. - **Molecular pathway:** mevalonate/isoprenoid biosynthesis → GGPP → protein geranylgeranylation; downstream RhoA/Rho-kinase and Rab-dependent vesicle trafficking. - **Cellular processes (GO):** protein geranylgeranylation (GO:0018344), isoprenoid biosynthetic process (GO:0008299), plasma-membrane repair (GO:0001778), vesicle-mediated transport (GO:0016192), autophagy (GO:0006914). - **Cell types (CL):** skeletal muscle fiber (CL:0000188), cochlear hair cell (CL:0000855), ovarian granulosa cell (CL:0000501), hepatocyte (CL:0000182). - **Subcellular compartments (GO CC):** cytoplasm/cytosol (GO:0005829), sarcomere Z-disc (GO:0030018), myofibril (GO:0030016), sarcolemma (GO:0042383), mitochondrion (GO:0005739). - **Chemical entities (CHEBI):** geranylgeranyl diphosphate (CHEBI:48861), isopentenyl diphosphate (CHEBI:128769), farnesyl diphosphate (CHEBI:175763).

Relationship to Perrault syndrome

Because sensorineural hearing loss + ovarian insufficiency is the definition of **Perrault syndrome**, *GGPS1* is now listed among Perrault-syndrome genes alongside a set that is otherwise dominated by mitochondrial/peroxisomal genes (HSD17B4, HARS2, CLPP, LARS2, TWNK, ERAL1, RMND1, DAP3, PRORP, MRPL50, MRPL49, MRPS7, PEX6, TFAM). *GGPS1* is mechanistically distinct — an isoprenoid-pathway gene rather than a mitochondrial-translation gene — and is uniquely accompanied by a **prominent muscular dystrophy**, which is the discriminating feature at the bedside.

Section-by-Section Disease Report

1. Disease Information

A rare Mendelian multisystem disorder: congenital-onset progressive proximal muscular dystrophy with variable sensorineural hearing loss and primary ovarian insufficiency. **Identifiers:** MONDO:0859189; OMIM #619518 (MDHLOV); MedGen C5561980; UMLS C5561980. **Synonyms:** MDHLO, MUDHLOV, "GGPS1-related/associated congenital muscular dystrophy," "GGPS1-associated muscular dystrophy with/without hearing loss." Information is derived from **aggregated disease-level resources** (OMIM/Orphanet/UniProt/ClinVar) built from a small number of published patient cohorts (≈22 patients total across three primary reports), not from EHR-scale data.

2. Etiology

Causal factor: purely genetic — biallelic hypomorphic missense variants in *GGPS1*. **Genetic risk factors:** the causal variants themselves (Finding 5); no established modifier genes. **Environmental risk/protective factors:** none identified — this is a monogenic disorder without known environmental modifiers. **Consanguinity** raises risk (homozygous alleles reported). **Gene-environment interactions:** none documented. Note: statins (HMG-CoA reductase inhibitors) reduce mevalonate-pathway flux upstream of GGPP and are theoretically of concern, but no clinical interaction data exist.

3. Phenotypes

See Finding 7 for the full HPO table with frequencies. **Obligate feature:** progressive proximal muscle weakness (HP:0003323), congenital onset (HP:0003577), elevated CK (HP:0003236). **Variable features:** sensorineural hearing loss (HP:0000407; ~46–91%), premature ovarian insufficiency (HP:0008209) in post-pubertal females, respiratory insufficiency (HP:0002093), short stature (HP:0004322), scoliosis (HP:0002650), loss of ambulation (HP:0002505). **Progression:** progressive; some patients wheelchair-bound by the second decade, others retain independent ambulation. **Quality-of-life impact:** substantial — mobility loss, respiratory compromise, deafness, and infertility.

4. Genetic/Molecular Information

Causal gene: *GGPS1* (HGNC:4249; gene MIM 606982; NM_004837.4). **Variants:** ultra-rare missense (Finding 5), clustered in the C-terminal catalytic domain, ACMG classifications ranging pathogenic → VUS. **Allele frequencies:** absent-to-ultra-rare in gnomAD v4 with no homozygotes (Finding 6). **Origin:** germline. **Functional consequence:** partial (hypomorphic) loss of enzyme function. **Modifier genes / epigenetics / chromosomal abnormalities:** none established for this disorder (large 1q42 CNVs in ClinVar are non-specific).

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious contributors are known. Disease is fully explained by biallelic *GGPS1* genotype.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full ordered causal chain, pathway/GO/CL/CHEBI annotations, and branch structure.

7. Anatomical Structures Affected

Organ level (primary): skeletal muscle (UBERON:0001134), esp. proximal limb-girdle muscles; **cochlea/inner ear** (UBERON:0001844); **ovary** (UBERON:0000992). **Secondary:** respiratory muscles/diaphragm → respiratory insufficiency; axial skeleton → scoliosis; **liver** (UBERON:0002107) in some. **Body systems:** musculoskeletal, auditory/sensory, endocrine/reproductive, respiratory. **Tissue/cell:** striated muscle fiber (CL:0000188), cochlear hair cells (CL:0000855), ovarian granulosa cells (CL:0000501). **Subcellular:** sarcomere Z-line/myofibril, sarcolemma, cytosol, mitochondria (enlarged on EM), autophagosomes. **Lateralization:** bilateral/symmetric.

8. Temporal Development

Onset: congenital (decreased fetal movement, weak cry, poor suck, motor delay). **Course:** chronic, progressive, lifelong. **Progression rate:** variable — wheelchair by second decade in severe cases vs. maintained ambulation in milder cases. Ovarian insufficiency manifests at expected puberty (primary amenorrhea / premature ovarian failure). **Critical periods:** perinatal and childhood for motor/respiratory decline; puberty for reproductive endocrine failure. No remission.

9. Inheritance and Population

Inheritance: autosomal recessive (HP:0000007). **Penetrance:** complete for myopathy; variable expressivity for hearing loss and ovarian insufficiency. **Epidemiology:** ultra-rare; no formal prevalence/incidence estimate (≈ 22 reported patients worldwide). **Founder/consanguinity:** homozygous alleles reported in consanguineous families; p.Arg261Gly is a recurrent (hotspot) allele. **Carrier frequency:** each pathogenic allele is absent-to-ultra-rare in gnomAD; no homozygotes observed. **Sex ratio:** both sexes affected by muscle/ear disease; ovarian insufficiency affects females only.

10. Diagnostics

Laboratory: elevated serum creatine kinase (near-universal). **Audiology:** confirms sensorineural hearing loss. **Endocrine:** elevated gonadotropins/low estradiol, primary amenorrhea in females (ovarian insufficiency). **Muscle biopsy/histopathology:** dystrophic changes, centrally nucleated fibers (HP:0003687), rimmed vacuoles (HP:0003805), autophagic material, enlarged mitochondria on EM. **Pulmonary function:** reduced forced vital capacity in some. **Genetic testing (definitive):** WES or WGS with shared-haplotype/segregation analysis identified the gene; targeted single-gene or Perrault-syndrome/congenital-muscular-dystrophy

panel testing of *GGPS1* (NM_004837.4) confirms diagnosis. CMA/karyotype not informative. **Differential diagnosis:** other Perrault-syndrome genes (mitochondrial-translation/peroxisomal — CLPP, LARS2, HARS2, TWNK, MRPL49, PRORP, etc.), other congenital/limb-girdle muscular dystrophies (notably dysferlinopathy and other membrane-repair myopathies), and other causes of primary ovarian insufficiency. Prominent congenital muscular dystrophy plus the *GGPS1* genotype distinguishes MDHLOV from classic Perrault syndrome.

11. Outcome / Prognosis

Chronic, progressive, lifelong disability. **Motor:** loss of ambulation in a subset (5/11 in the original cohort). **Respiratory:** insufficiency in the majority — the principal driver of morbidity/mortality. **Reproductive:** infertility from ovarian insufficiency. **Sensory:** permanent hearing loss. No formal survival statistics; prognosis is dominated by respiratory muscle involvement and mobility loss. **Prognostic factors:** severity/onset of weakness and respiratory decline; genotype-phenotype correlation is suggested but not firmly established given small numbers.

12. Treatment

No disease-specific/curative therapy exists. Management is **supportive and multidisciplinary:** physical/occupational therapy and mobility aids; respiratory monitoring and non-invasive ventilation for respiratory insufficiency; hearing aids or cochlear implantation for sensorineural hearing loss; **hormone replacement therapy** for primary ovarian insufficiency (estrogen/progesterone; bone-health protection); orthopedic management (scoliosis, contractures); nutritional support for failure to thrive; genetic counseling. **Advanced/experimental therapeutics:** none approved; no gene, cell, or RNA therapies in trials for this disorder. Theoretically, substrate-supplementation or pathway-modulation strategies are of mechanistic interest but untested. **Pharmacogenomic caution:** statins (which lower mevalonate flux upstream of GGPP) are theoretically undesirable but have no specific evidence base here. Suggested NCIT intervention terms: physical therapy (NCIT:C15327), hormone replacement therapy (NCIT:C62556), mechanical ventilation (NCIT:C70909), cochlear implant (NCIT:C99913).

13. Prevention

No primary prevention (monogenic). **Genetic counseling** for at-risk families (25% recurrence risk per pregnancy for carrier couples). **Carrier and cascade testing** of relatives; **prenatal diagnosis** and **preimplantation genetic testing** are options once the familial variants are known. Newborn screening does not cover this disorder. Tertiary prevention = managing complications (respiratory support, HRT for bone health, scoliosis management).

14. Other Species / Natural Disease

No naturally occurring *GGPS1* disease is reported in companion animals or wildlife (no OMIA entry described). Orthologs are highly conserved (Finding 9): mouse *Ggps1* (14593), rat (291211), zebrafish (336798), *Xenopus* (549876), *Drosophila* (38816), yeast *BTS1* (856036). No zoonotic potential (non-infectious genetic disease).

15. Model Organisms

Disease-specific model: a **p.Tyr259Cys knock-in mouse** (Foley et al., 2020) recapitulates dystrophic muscle histology (autophagic material, enlarged mitochondria) — good recapitulation of the muscle phenotype; hearing/ovarian phenotypes less characterized. **Conditional deletion mice:** skeletal-muscle-specific *Ggps* knockout (insulin resistance via RhoA prenylation; PMID: 26112408) and liver-specific knockout (adipose remodeling via Rab27A; PMID: 32024826) — model gene function but not the exact human disease. **Limitation:** global null is lethal, so only hypomorphic/conditional models are informative. Model resources: MGI, IMPC, IMSR. Complementary in-vitro model: **patient-derived myogenic cells** showing the membrane-repair defect.

Evidence Base

PMID	Title (abbrev.)	Role in this report
32403198	<i>GGPS1 Mutations Cause MDHLOV Syndrome</i> (Foley 2020)	Landmark gene-discovery paper. Supports Findings 1, 3, 8, 9 — WES/haplotype discovery, clinical triad, enzyme function, delayed membrane repair in patient cells, Y259C knock-in mouse.
35869884	<i>GGPS1-associated MD with and without hearing loss</i> (Kaiyrzhanov 2022)	Supports Findings 2, 5 — 11 more patients; hearing loss in only 46%; establishes variable expressivity and "ultra-rare biallelic" nature.
38129970	<i>Expanding the phenotypic/genotypic spectrum</i> (Altassan 2024)	Supports Finding 2 — isolated proximal weakness + elevated transaminases with spared hearing; adds hepatic involvement.
12736685	<i>Defective membrane repair in dysferlin-deficient MD</i> (Bansal 2003)	Supports Finding 3 — establishes active Ca ²⁺ -dependent sarcolemmal repair as a disease-relevant muscle process.
23663983	<i>Oligomerization of rab/effector complexes</i>	Supports Finding 3 — Rab GTPases require C-terminal prenylation for membrane tethering; explains downstream defect.
26112408	<i>Lipid-induced muscle insulin resistance via GGPPS/RhoA</i>	Supports Finding 9 — skeletal-muscle <i>Ggpps</i> conditional KO; RhoA geranylgeranylation.
32024826	<i>Liver governs adipose remodelling via EVs</i>	Supports Finding 9 — liver <i>Ggpps</i> KO; Rab27A geranylgeranylation controls EV secretion.
31427080	<i>Crystal structure of GGPP synthase (crtE)</i>	Supports Finding 8 — homohexameric (trimer-of-dimers) architecture of GGPP synthases.
42283975	<i>Comprehensive insights into Perrault syndrome</i>	Context — places <i>GGPS1</i> among 15 Perrault-syndrome genes; clinical/genetic heterogeneity.
24784578 , 27286750 , 32087766 , 26911675	Dysferlin/Annexin-A5/AMPK/ANO5 membrane-repair studies	Context — mechanistic framework for vesicle-fusion-based sarcolemmal repair, the process inferred to fail here.
36116551 , 40325959	Rab prenylation/localization; Rep-	

PMID	Title (abbrev.)	Role in this report
	deficiency retinal degeneration	Context — consequences of failed Rab geranylgeranylation (membrane mislocalization, cell death).

Limitations and Knowledge Gaps

- 1. Small evidence base.** The entire disease description rests on ≈ 22 patients across three primary reports; prevalence, incidence, survival, and robust genotype-phenotype correlations cannot be estimated.
- 2. Inferred (not demonstrated) extra-muscular mechanism.** The membrane-repair defect is directly shown only in muscle. The cochlear and ovarian branches are mechanistic inferences from the shared prenylation defect; the specific under-prenylated GTPases and cellular events in inner ear and ovary have not been experimentally defined.
- 3. Which Rab/Rho substrates are limiting?** Direct proteomic/prenylomic evidence identifying the specific hypoprenylated small GTPases in patient tissues is lacking.
- 4. VUS burden.** Several *GGPS1* alleles (p.Gly255Asp, p.Leu264Val, p.Leu182Pro) remain VUS; functional enzymatic assays are needed for reclassification.
- 5. No therapeutic data.** No trials, no natural-history registry, and no validated biomarkers for progression beyond CK.
- 6. Model gaps.** Knock-in mouse muscle phenotype is described, but auditory and ovarian phenotypes in the model are not fully characterized; global null lethality limits modeling of complete deficiency.

Proposed Follow-up Experiments / Actions

- 1. Prenylomics in patient cells/tissues** — quantify unprenylated Rab/Rho GTPases (e.g., in-vitro prenylation with biotin-geranylgeranyl) in patient myoblasts, and if accessible, cochlear/granulosa-cell models, to directly test step 4 of the causal chain.
- 2. Phenotype the Y259C knock-in mouse for hearing and fertility** — ABR audiometry and ovarian histology/reproductive assays to test the cochlear and ovarian branches in vivo.

3. **iPSC-derived organoids** — patient-derived iPSC inner-ear organoids and ovarian/granulosa models to interrogate tissue-specific trafficking defects.
4. **Functional reclassification of VUS** — express VUS alleles and measure GGPP-synthase catalytic activity/thermostability to move VUS toward pathogenic/benign.
5. **GGPP/mevalonate-pathway rescue screens** — test whether GGPP supplementation, geranylgeraniol, or upstream pathway modulation rescues membrane-repair kinetics in patient myogenic cells (proof-of-concept for a metabolic therapy).
6. **International registry** — establish a natural-history registry (via Perrault-syndrome and congenital-muscular-dystrophy networks) to capture prevalence, progression, respiratory outcomes, and genotype-phenotype correlations.
7. **Structural modeling of hotspot residues** — map Phe257/Tyr259/Arg261/Val285 onto the GGPP-synthase active site (homology/AlphaFold) to explain why Arg261 is a mutational hotspot.

Report compiled from 10 confirmed findings and 25 reviewed publications over 5 investigation iterations. Evidence types are human clinical (patient cohorts), model organism (mouse conditional/knock-in), and in vitro (patient-derived myogenic cells; bacterial/structural enzymology).
